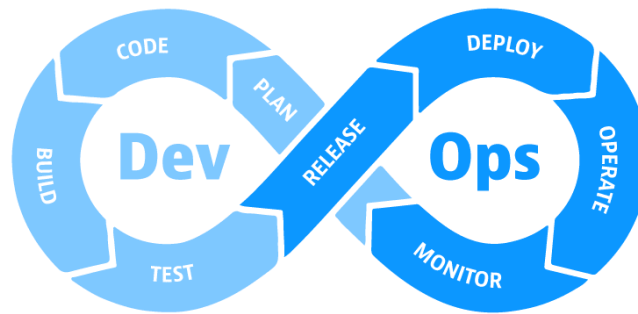


# DevOps

## 1. What is DevOps?

Think of DevOps as a bridge that connects two groups of people who used to work in separate rooms and rarely talked to each other: the **Developers** (the people who write the code) and **Operations** (the people who run the servers and maintain the software).



- **Better Teamwork:** It breaks down the walls so programmers and system administrators work together from day one.
- **Automation First:** It uses tools to do repetitive tasks automatically, like checking the code for errors or putting it online.
- **Continuous Cycle:** Instead of building software all at once, teams constantly write, test, and release small updates over and over again.
- **Early Warnings:** It uses smart monitoring tools to instantly alert the team if something breaks or runs slowly for real users.

## 2. What are the Advantages of DevOps?

- **Super Fast Releases:** Companies can launch new app features or fixes in hours or days, rather than waiting months.
- **Fewer Crashes:** Because updates are broken down into small pieces and tested by computers, they are much less likely to break the system.
- **Quick Fixes:** If a bug does get through and the app crashes, DevOps tools allow the team to fix it or undo the update in minutes.
- **Less Manual Toil:** Automation takes care of boring, repetitive tasks, giving engineers more time to build fun new features.

## 3. What is Agile Methodology?

Agile is a mindset for managing projects. Instead of trying to plan out a massive 2-year project perfectly on paper (which usually fails), Agile focuses on being flexible and learning as you go.



- **Small Slices:** It breaks down a big project into tiny, workable pieces that can be finished quickly.
- **Flexibility:** If the customer changes their mind about a feature halfway through, Agile makes it easy to switch directions without losing progress.
- **People Over Rules:** It values real human conversations and collaboration over rigid rules and heavy paperwork.
- **Step-by-Step Growth:** The team builds a very basic version of the software first, gathers feedback, and improves it over time.

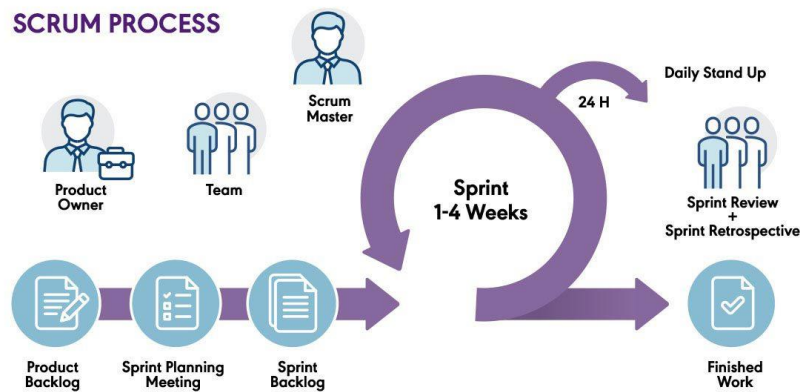
#### 4. What are the Advantages of Agile?

- **Happier Customers:** Customers get to see and test parts of the product early on, ensuring they get exactly what they want.
- **Less Financial Risk:** If a project isn't working out, the team finds out in a few weeks rather than a year later, saving lots of money.
- **High Quality:** Testing happens constantly throughout the project, meaning bugs are caught and fixed early before they cause big issues.
- **No Surprises:** Everyone—from the developers to the business bosses—knows exactly what is being worked on at any given moment.

#### 5. What is Scrum?

Scrum is the most popular "recipe" or framework used to put Agile into practice. It gives a team a clear set of roles and rules to follow.

- **Sprints:** Work is divided into short, fixed time frames called sprints, which usually last 2 to 4 weeks.
- **The Backlog:** A big, ordered "To-Do List" of every single feature, tweak, or bug fix needed for the software.



- **Three Simple Roles:**
  - **Product Owner:** The person who decides *what* needs to be built based on what customers want.
  - **Scrum Master:** A coach who helps the team follow Scrum rules and clears away anything blocking their progress.
  - **Developers:** The crew (programmers, designers, testers) who actually build the product.

## 6. What are the Important Meetings in Scrum?

Scrum teams have four key meetings during every single sprint to keep everyone on track.

- **Sprint Planning (At the Start):** The team looks at the master To-Do list, chooses what they can realistically finish in the next 2–4 weeks, and makes a plan to do it.
- **Daily Stand-up (Every Morning):** A quick, 15-minute meeting where everyone answers three things: *What did I do yesterday? What am I doing today? Is anything blocking me?*
- **Sprint Review (At the End):** A show-and-tell session where the team demonstrates the working features they just built to the stakeholders and gets their thoughts.
- **Sprint Retrospective (After the Review):** An internal team meeting to look back at how they worked together and pick 1 or 2 ways to make the next sprint even better.